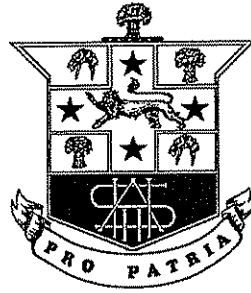


# HURLSTONE AGRICULTURAL HIGH SCHOOL



## MATHEMATICS EXTENSION 1 2009 YEAR 12 TRIAL HSC EXAMINATION

EXAMINERS ~ H. CAVANAGH, D CRANCHER, J DILLON, G HUXLEY.

### GENERAL INSTRUCTIONS

- |                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                        |
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| <ul style="list-style-type: none"><li>• Reading Time – 5 minutes.</li><li>• Working Time – 2 hours.</li><li>• Attempt <b>all</b> questions.</li><li>• Questions are of equal value.</li><li>• <b>All</b> necessary working should be shown in every question.</li><li>• This paper contains seven (7) questions.</li></ul> | <ul style="list-style-type: none"><li>• Marks may not be awarded for careless or badly arranged work.</li><li>• Board approved calculators and MathAids may be used.</li><li>• <b>Each question is to be started in a new answer booklet.</b> Additional booklets are available if required.</li><li>• This examination must <b>NOT</b> be removed from the examination room</li></ul> |
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STUDENT NAME: \_\_\_\_\_

STUDENT NUMBER: \_\_\_\_\_

**QUESTION 1** (12 marks) Start a new booklet

**Marks**

- (a) Find the acute angle, to the nearest degree, between the lines

**2**

$$y = 5 - x \text{ and } y = \frac{3}{2}x + 5.$$

- (b) (i) Show that  $(x-2)$  is a factor of  $x^3 - 3x^2 + 4$

**1**

- (ii) Express  $x^3 - 3x^2 + 4$  as a product of three linear factors

**2**

- (c) If  $y = \log_{\frac{1}{a}}\left(\frac{1}{N}\right)$ , where  $a > 0$  and  $N > 0$ , show that  $y = \log_a N$

**2**

- (d) The point  $(0, 4)$  divides the interval from  $(a, b)$  to  $(b, a)$  internally in the ratio 3: 1.  
Find the values of  $a$  and  $b$ .

**3**

- (e) Find the Cartesian equation of the locus of a point  $P(x, y)$

**2**

$$\text{where } x = 2 \cos \theta \text{ and } y = \frac{1}{2} \sin \theta$$

**QUESTION 2** (12 marks) Start a new booklet**Marks**

(a) Find  $\lim_{x \rightarrow 0} \frac{\sin x}{5x}$  **1**

(b) A particle moves in a straight line so that its velocity,  $v$ , in minutes per second at time  $t$  is given by  $v = 4 - 2t$ . Initially, the particle is at  $x = 1$ .

(i) Find the displacement  $x$  of the particle as a function of  $t$ . **1**

(ii) When is the particle at rest and what is its acceleration at that time? **2**

(c) Heat is applied to a metallic disc for  $t$  seconds.  
Its area,  $A \text{ cm}^2$ , increases at a rate given by:

$$\frac{dA}{dt} = t^2 - 2t + 1$$

(i) Find the rate at which the area of the metallic disc is increasing at the end of the third second. **1**

(ii) Before heat is applied the area of the metallic disc is  $10 \text{ cm}^2$ . **2**  
Heat is then applied to the metallic disc for 3 seconds.  
What is the area of this disc at the end of three seconds?

(d) Given that  $0 < x < \frac{\pi}{4}$  prove that :  $\tan\left(\frac{\pi}{4} + x\right) = \frac{\cos x + \sin x}{\cos x - \sin x}$  **3**

(e) Let  $f(x) = \log_e [(x+2)(x-4)]$  **2**  
What is the domain of  $f(x)$ ?

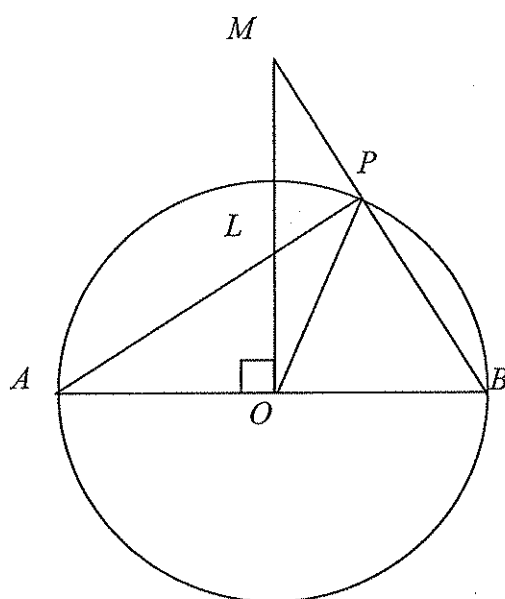
**QUESTION 3** (12 marks) Start a new booklet

**Marks**

- (a) In how many ways can the letters of the word **SUCCESS** be arranged? 2  
All letters must be used.

- (b) The staff in an office consists of 4 males and 7 females. 2  
What is the probability that a randomly chosen committee of 5 staff can be formed which contains exactly 3 females?

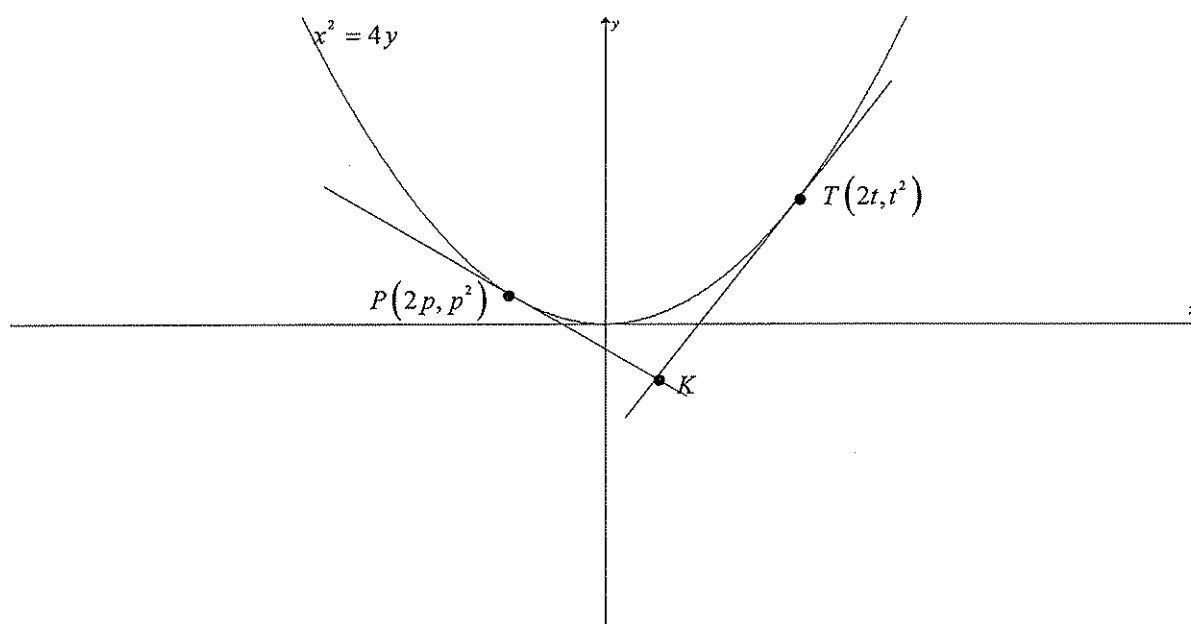
(c)



$O$  is the centre of the circle  $ABP$ .  $MO \perp AB$ .  $M, P$  and  $B$  are collinear.  
 $MO$  intersects  $AP$  at  $L$ .

- (i) Prove that  $A, O, P$  and  $M$  are concyclic. 2
- (ii) Prove that  $\angle OPA = \angle OMB$ . 2
- (d) (i) Show that  $P(x) = x^3 - x^2 - x - 1$  has a zero between 1 and 2. 1
- (ii) Take  $x = 2$  as a first approximation and use Newton's method to calculate a second approximation. 2
- (iii) Explain why  $x = 1$  was not a suitable first approximation in this case. 1

(a)



The diagram shows the graph of the parabola  $x^2 = 4y$  and also the tangents at  $T(2t, t^2)$  and  $P(2p, p^2)$ . The tangents intersect at  $K$ .

(i) Prove that the equation of the tangent at  $T$  is  $y = tx - t^2$  **2**

(ii) Show that the coordinates of the point  $K$ , the point where the tangents at  $T$  and  $P$  intersect, are  $(p+t, pt)$ . **2**

(iii) The angle  $TKP$  is a right angle. **1**  
Show that the locus of  $K$  is a straight line.

(b) Solve the inequation:  $\frac{x}{x-3} < 4$  **3**

(c) The polynomial  $P(x) = 6x^3 - 7x^2 + ax + b$  has a zero at  $x = -1$ .  
The remaining zeros are reciprocals.

(i) By examining the product of the three roots, determine the values of  $b$ ,  
and hence of  $a$ . **2**

(ii) Find all zeros of  $P(x)$ . **2**

**QUESTION 5** (12 marks) Start a new booklet

**Marks**

- (a) Use mathematical induction to prove that, for every positive integer  $n$ ,  $13 \times 6^n + 2$  is divisible by 5. **3**
- (b) Differentiate with respect to  $x$ :  $x e^{\sin x}$  **2**
- (c) The curve  $y = \sin 2x$  and the line  $y = \frac{4x}{\pi}$  intersect at  $x = 0$  and  $x = \frac{\pi}{4}$ . **2**  
Find the area bounded by the curve  $y = \sin 2x$  and the line  $y = \frac{4x}{\pi}$  in the first quadrant. (Answer in terms of  $\pi$ ).
- (d) Find the exact volume of the solid formed when the area between the curve  $y = \log_e x$  and the  $y$ -axis is rotated about the  $y$ -axis between  $y = 0$  and  $y = 3$ . **2**
- (e) Use the substitution  $u = \sqrt{x}$  to evaluate  $\int_0^3 \frac{dx}{\sqrt{x}(1+x)}$  **3**

**QUESTION 6** (12 marks) Start a new booklet**Marks**

(a) Find the exact value of  $\int_0^1 \frac{dx}{\sqrt{2-x^2}}$ . **2**

(b) State the domain and range of the function  $y = 2 \sin^{-1}(3x)$ . **3**  
Sketch the graph.

(c) If  $y = \cos^{-1}\left(\frac{1}{x}\right)$  for  $0 \leq y \leq \frac{\pi}{2}$ , show that:  $\frac{dy}{dx} = \frac{1}{x\sqrt{x^2-1}}$  **2**

(d) Consider the function  $f(x) = e^{-x} - e^x$

(i) Show that  $f(x)$  is decreasing for all values of  $x$ . **1**

(ii) Show that the inverse function is given by

$$f^{-1}(x) = \log_e \left( \frac{\sqrt{x^2 + 4} - x}{2} \right).$$
**3**

(iii) Hence, or otherwise, solve  $e^{-x} - e^x = 6$  **1**

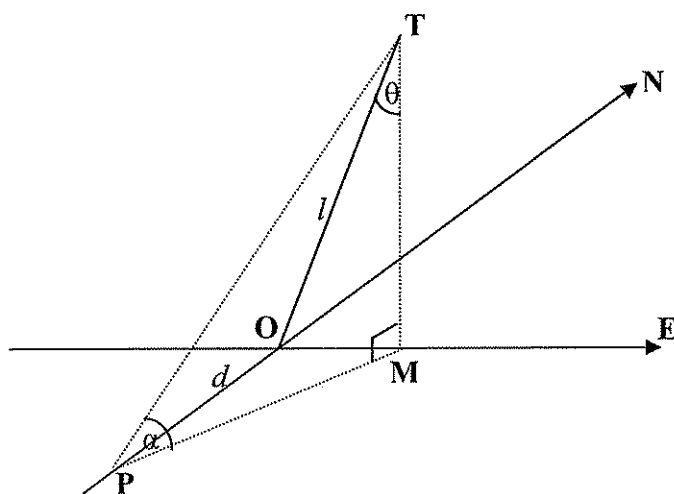
Give your answer correct to two decimal places.

(a) (i) Show that  $\int_0^{\frac{\pi}{4}} \cos^2 x \, dx = \frac{\pi+2}{8}$  2

(ii) The region under the curve  $y = \cos x + \sec x$ , above the  $x$ -axis, between  $x = 0$  and  $x = \frac{\pi}{4}$ , makes a revolution about the  $x$ -axis. 3

Show that the volume traced out is  $\frac{5\pi(\pi+2)}{8}$  units<sup>3</sup>.

(b)



A pole,  $OT$ , of length  $l$  metres, stands on horizontal ground.

The pole leans towards the east, making an angle of  $\theta$  with the vertical.

From  $P$ ,  $d$  metres south of  $O$ , the elevation of  $T$  is  $\alpha$

(i) Find expressions, in terms of  $l$  and  $\theta$ , for  $OM$  and  $MT$ . 2

(ii) Prove that  $PM = l \cos \theta \cot \alpha$ . 2

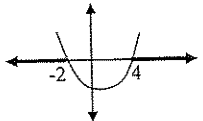
(iii) Prove that  $l^2 = \frac{d^2}{\cos^2 \theta \cot^2 \alpha - \sin^2 \theta}$ . 2

(iv) Find the length of the pole, to the nearest metre, if  $d = 25$ ,  $\theta = 20^\circ$ ,  $\alpha = 24^\circ$ . 1

| Year 12 Mathematics Extension 1 Trial HSC 2009                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                  |                                                                                                                                                                    |                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Question No. 1                                                                                                                                                                                                                                                                                                        | Solutions and Marking Guidelines                                                                                                                                                                                                                                 |                                                                                                                                                                    |                                                                                                                                                                                                                                 |
| <b>Outcomes Addressed in this Question</b><br>PE3: solves problems involving polynomials and parametric representations.<br>H3: manipulates algebraic expressions involving logarithmic and exponential functions<br>HS: applies appropriate techniques from the study of geometry and trigonometry to solve problems |                                                                                                                                                                                                                                                                  |                                                                                                                                                                    |                                                                                                                                                                                                                                 |
| Outcome                                                                                                                                                                                                                                                                                                               | Solutions                                                                                                                                                                                                                                                        | Marking Guidelines                                                                                                                                                 |                                                                                                                                                                                                                                 |
| Q1(a)<br>HS                                                                                                                                                                                                                                                                                                           | $y = 5 - x$ and $y = \frac{3}{2}x + 5$<br>$m_1 = -1$ $m_2 = \frac{3}{2}$<br>$\tan \theta = \left  \frac{m_2 - m_1}{1 + m_1 m_2} \right $<br>$= \left  \frac{\frac{3}{2} + 1}{1 - \frac{3}{2}} \right $<br>$= 5$<br>$\theta = 79^\circ$                           | 2 marks: Fully correct solution<br>1 mark: Attempts application of formula with correct gradients.<br>OR arithmetic error in gradients, correct working for angle. |                                                                                                                                                                                                                                 |
| Q1(b)(i)<br>PE3                                                                                                                                                                                                                                                                                                       | $(x - 2)$ is a factor of $P(x) = x^3 - 3x^2 + 4$ if $P(2) = 0$<br>$P(2) = 2^3 - 3(2)^2 + 4$<br>$= 0$<br>$\therefore (x - 2)$ is a factor.                                                                                                                        | 1 mark<br>Correct use of factor theorem to prove the given result, or correct division.                                                                            |                                                                                                                                                                                                                                 |
| Q1(b)(ii)<br>PE3                                                                                                                                                                                                                                                                                                      | $\begin{array}{r} x^2 - x - 2 \\ x-2 \overline{) x^3 - 3x^2 + 4} \\ \underline{x^3 - 2x^2} \phantom{+ 4} \\ -x^2 \phantom{+ 4} \\ \underline{-x^2 + 2x} \phantom{+ 4} \\ -2x + 4 \\ \underline{-2x + 4} \\ \hline \end{array}$                                   | $\therefore P(x) = (x - 2)(x^2 - x - 2)$<br>$= (x - 2)(x + 1)(x - 2)$<br>$= (x - 2)^2(x + 1)$                                                                      | 2 marks<br>Correctly divides the given polynomial, and factorises the resulting quotient to show the 3 linear factors. OR<br>Factorises correctly by inspection.<br>1 mark<br>Substantial progress towards the correct solution |
| Q1(c)<br>HS                                                                                                                                                                                                                                                                                                           | $y = \log_{\frac{1}{a}} \left( \frac{1}{N} \right)$<br>$\left( \frac{1}{a} \right)^y = \frac{1}{N}$<br>Take reciprocals<br>$a^y = N$<br>$\therefore y = \log_a N$                                                                                                | 2 marks: Fully correct solution<br>1 mark: Sufficient progress towards correct result                                                                              |                                                                                                                                                                                                                                 |
| Q1(d)<br>PE3                                                                                                                                                                                                                                                                                                          | At $(0, 4)$ , $\begin{cases} a + 3b = 0 \\ b + 3a = 16 \end{cases}$ Solving simultaneously, $a = 6, b = -2$                                                                                                                                                      | 3 marks<br>Correctly produces 2 equations, and correctly solves simultaneously.                                                                                    |                                                                                                                                                                                                                                 |
| Q1(e)<br>PE3                                                                                                                                                                                                                                                                                                          | $\begin{cases} x = 2 \cos \theta \\ y = \frac{1}{2} \sin \theta \end{cases} \rightarrow \begin{cases} \cos \theta = \frac{x}{2} \\ \sin \theta = 2y \end{cases}$<br>Since $\cos^2 \theta + \sin^2 \theta = 1$<br>$\frac{x^2}{4} + 4y^2 = 1$<br>$x^2 + 16y^2 = 4$ | 2 marks: Fully correct solution<br>1 mark: Some progress towards correct result, including use of the Pythagorean result.                                          |                                                                                                                                                                                                                                 |

| Year 12                             | Extension 1 Mathematics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Trial HSC 2009                                                                                                                                                   |
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| Question No. 3                      | Solutions and Marking Guidelines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                  |
| Outcomes Addressed in this Question |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                  |
| PE3                                 | solves problems involving permutations and combinations, inequalities, polynomials, circle geometry and parametric representations                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                  |
|                                     | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marking Guidelines                                                                                                                                               |
| (a)                                 | Number of ways = $\frac{7!}{3!2!} = 420$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Award 2 for correct solution.<br><br>Award 1 for attempting to use an appropriate process.                                                                       |
| (b)                                 | Number of unrestricted committees of 5 = $\binom{11}{5}$<br><br>Number of committees of 3 females = $\binom{7}{3}\binom{4}{2}$<br><br>$\therefore \text{Probability} = \frac{\binom{7}{3}\binom{4}{2}}{\binom{11}{5}} = \frac{5}{11}$                                                                                                                                                                                                                                                                                                                                               | Award 2 for correct solution.<br><br>Award 1 for attempting to use an appropriate process.                                                                       |
| (c)                                 | (i) Let $\angle PAO = \theta$<br>$\angle APB = 90^\circ$ (angle in a semi-circle)<br>$\therefore \angle ABP = (90 - \theta)^\circ$ (angle sum of $\triangle ABP$ )<br>$\therefore \angle BMO = \theta$ (angle sum of $\triangle BMO$ )<br>$\therefore \angle OAP = \angle BMO$<br>$\therefore AOPM$ is a cyclic quadrilateral ( $OP$ subtends equal angles at $A$ and $M$ )<br>$\therefore A, O, P$ , and $M$ are concyclic<br><br>(ii) $\angle OPA = \theta$ (angles opposite equal sides $OP$ and $OA$ )<br>$\angle BMO = \theta$ (above)<br>$\therefore \angle OPA = \angle OMB$ | Award 2 for correct solution.<br><br>Award 1 for substantial progress towards solution.                                                                          |
| (d)                                 | (i) $P(1) = 1 - 1 - 1 - 1 = -2$<br>$P(2) = 8 - 4 - 2 - 1 = 1$<br>$\therefore$ Since $P(x)$ is a continuous function and the sign changes, there must be a root between 1 and 2.<br><br>(ii) $P'(x) = 3x^2 - 2x - 1$<br>$P(2) = 1$ and $P'(2) = 7$<br>$\therefore x_2 = 2 - \frac{1}{7} = 1\frac{6}{7}$<br><br>(iii) $P'(1) = 0$<br>This means that $x = 1$ is a stationary point<br>$\therefore$ Newton's method will fail.                                                                                                                                                         | Award 1 for correct solution.<br><br>Award 2 for correct solution.<br><br>Award 1 for substantial progress towards solution.<br><br>Award 1 for correct solution |

| Year 12 Mathematics Extension 1 Task 4 Trial HSC 2009                                                                          |                                                                                                                                                                                                                             |                                                                                                  |
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| Question No. 2 Solutions and Marking Guidelines                                                                                |                                                                                                                                                                                                                             |                                                                                                  |
| Outcomes Addressed in this Question                                                                                            |                                                                                                                                                                                                                             |                                                                                                  |
| H5 applies appropriate techniques from the study of calculus, geometry, probability, trigonometry and series to solve problems |                                                                                                                                                                                                                             |                                                                                                  |
| Outcome                                                                                                                        | Solutions                                                                                                                                                                                                                   | Marking Guidelines                                                                               |
| Q2.                                                                                                                            |                                                                                                                                                                                                                             |                                                                                                  |
| H5                                                                                                                             | (a) $\frac{1}{5}$                                                                                                                                                                                                           | 1 mark correct answer                                                                            |
| H5                                                                                                                             | (b) (i) $x = \int (4 - 2t) dt$<br>$= 4t - t^2 + C$<br>when $x = 1, t = 0 \Rightarrow C = 1$<br>$x = 4t - t^2 + 1$                                                                                                           | 1 mark correct answer                                                                            |
|                                                                                                                                | (ii) The particle is at rest when $v = 0$<br>$0 = 4 - 2t$<br>$0 = 2(2 - t)$<br>$t = 2$<br>$a = \frac{dv}{dt} = -2$<br><br>$\therefore$ The particle is at rest after 2 seconds and has constant acceleration of $-2m/s^2$ . | 1 mark for correct time particle is at rest<br><br>1 mark for correct acceleration at that time. |
| H5                                                                                                                             | (c) (i) At $t = 3$<br>$\frac{dA}{dt} = (3)^2 - 2 \cdot (3) + 1$<br>$= 4$<br>The rate at which the disc is increasing is $4cm^2/s$                                                                                           | 1 mark for correct answer                                                                        |

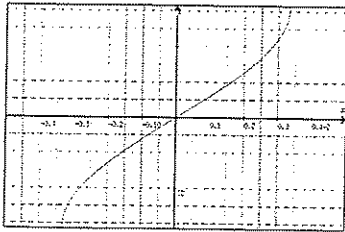
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|    | (ii) $A = \int (t^2 - 2t + 1) dt$<br>$= \frac{t^3}{3} - t^2 + t + C$<br>When $t = 0, A = 10 \Rightarrow C = 10$<br>$A = \frac{1}{3}t^3 - t^2 + t + 10$<br>At $t = 3, A = \frac{1}{3}(3)^3 - (3)^2 + 3 + 10$<br>$= 13cm^2$                                                                                                                                                                                                                                                                                                                                                                                                 | 1 mark awarded for partial correct solution<br><br>2 marks awarded for complete correct solution                                                             |
| H5 | (d) $LHS = \tan\left(\frac{\pi}{4} + x\right)$<br>$= \frac{\tan \frac{\pi}{4} + \tan x}{1 - \tan \frac{\pi}{4} \tan x} \left[ \text{since, } \tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} \right]$<br>$= \frac{1 + \tan x}{1 - \tan x}$<br>$= \frac{\frac{\cos x}{\cos x} + \frac{\sin x}{\cos x}}{\frac{\cos x}{\cos x} - \frac{\sin x}{\cos x}} \left[ \text{since, } \tan x = \frac{\sin x}{\cos x} \text{ \& } 1 = \frac{\cos x}{\cos x} \right]$<br>$= \frac{\cos x + \sin x}{\cos x - \sin x}$<br>$= RHS$<br><br>$\therefore \tan\left(\frac{\pi}{4} + x\right) = \frac{\cos x + \sin x}{\cos x - \sin x}$ | 1 mark awarded for partial correct solution<br><br>2 marks awarded for further partial correct solution<br><br>3 marks awarded for complete correct solution |
| H5 | (e) $f(x) = \log_v[(x+2)(x-4)]$<br>$(x+2)(x-4) > 0$<br>$\therefore x < -2, x > 4$<br>                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1 mark awarded for partial correct solution<br><br>2 marks awarded for complete correct solution                                                             |

| Year 12<br>Question No. 4                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Extension 1 Mathematics<br>Solutions and Marking Guidelines                                                                                                                                                                          | Trial HSC 2009 |
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| Outcomes Addressed in this Question                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                      |                |
| PE3 solves problems involving permutations and combinations, inequalities, polynomials, circle geometry and parametric representations |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                      |                |
| Solutions                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Marking Guidelines                                                                                                                                                                                                                   |                |
| (a)                                                                                                                                    | <p>(i) <math>y = \frac{x^2}{4}</math><br/> <math>y' = \frac{x}{2}</math><br/> At <math>(2t, t^2)</math> <math>y' = t = m_{\text{tangent}}</math><br/> <math>\therefore</math> Equation of tangent is<br/> <math>y - t^2 = t(x - 2t) = tx - 2t^2</math><br/> <math>\therefore y = tx - t^2</math></p> <p>(ii) Equation of tangent at <math>P</math> is<br/> <math>y = px - p^2</math><br/> <math>\therefore</math> At point of intersection<br/> <math>px - p^2 = tx - t^2</math><br/> <math>(p - t)x = p^2 - t^2 = (p - t)(p + t)</math><br/> <math>x = p + t</math> (<math>\because p \neq t</math>)<br/> Sub. into <math>y = px - p^2</math><br/> <math>y = p(p + t) - p^2 = pt</math><br/> <math>\therefore</math> Intersect @ <math>(p + t, pt)</math></p> <p>(iii) If <math>\angle TKP = 90^\circ</math><br/> <math>m_{TK} \times m_{PK} = -1</math><br/> <math>\therefore t \times p = -1 \Rightarrow pt = -1</math><br/> <math>\therefore</math> Locus of <math>K</math> is <math>x = p + t, y = -1</math><br/> <math>\therefore</math> Locus of <math>K</math> is the straight line <math>y = -1</math>.</p> | <p>Award 2 for correct solution.<br/> Award 1 for attempting to use an appropriate process.</p> <p>Award 2 for correct solution.<br/> Award 1 for attempting to use an appropriate process.</p> <p>Award 1 for correct solution.</p> |                |
| (b)                                                                                                                                    | $\frac{x}{x-3} < 4$<br>$x(x-3) < 4(x-3)^2$<br>$(x-3)(x-4(x-3)) < 0$<br>$(x-3)(12-3x) < 0$<br>$(x-3)(4-x) < 0$<br>$\therefore x < 3$ or $x > 4$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>Award 3 for correct solution.<br/> Award 2 for substantial progress towards solution.<br/> Award 1 for limited progress towards solution</p>                                                                                      |                |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                           |
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| (c) | <p>(i) <math>P(-1) = 0 \Rightarrow 6(-1)^3 - 7(-1)^2 + (-1)a + b = 0</math><br/> <math>\therefore b - a = 13</math> (1)<br/> Roots are <math>-1, \alpha, \frac{1}{\alpha} \Rightarrow -1 \times \alpha \times \frac{1}{\alpha} = -\frac{b}{6}</math><br/> <math>\therefore \frac{b}{6} = 1 \Rightarrow b = 6</math><br/> Substitute into (1) <math>\Rightarrow a = -7</math></p> <p>(ii) <math>P(x) = 6x^3 - 7x^2 - 7x + 6</math><br/> <math>-1 + \alpha + \frac{1}{\alpha} = -\frac{-7}{6} = \frac{7}{6}</math><br/> <math>\alpha + \frac{1}{\alpha} = \frac{13}{6}</math><br/> <math>\alpha^2 + 1 = \frac{13}{6}\alpha</math><br/> <math>6\alpha^2 - 13\alpha + 6 = 0</math><br/> <math>(3\alpha - 2)(2\alpha - 3) = 0</math><br/> <math>\therefore \alpha = \frac{2}{3}</math> or <math>\alpha = \frac{3}{2}</math><br/> <math>\therefore</math> Zeros are <math>-1, \frac{2}{3}, \frac{3}{2}</math>.</p> | <p>Award 2 for correct solution.<br/> Award 1 for substantial progress towards solution.</p> <p>Award 2 for correct solution.<br/> Award 1 for substantial progress towards solution.</p> |
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| Year 12 Mathematics Extension 1 Task 4 Trial HSC 2009 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                             |
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| Question No. 5 Solutions and Marking Guidelines       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                             |
| Outcomes Addressed in this Question                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                             |
| P7                                                    | determines the derivative of a function through routine application of the rules of differentiation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                             |
| PE2                                                   | uses multi-step deductive reasoning in a variety of contexts                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                             |
| H8                                                    | uses techniques of integration to calculate areas and volumes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                             |
| HE6                                                   | determines integrals by reduction to a standard form through a given substitution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                             |
| Outcome                                               | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marking Guidelines                                                                                                                                          |
| PE2                                                   | <p>5.<br/>(a)<br/>Let <math>S_n: 13 \times 6^n + 2</math> is divisible by 5<br/>i.e. <math>S_n: 13 \times 6^n + 2 = 5K</math> where <math>K, n</math> are positive integers.<br/>For <math>n=1</math>, <math>LHS = 13 \times 6^1 + 2 = 80</math><br/><math>RHS = 5K = 5 \times 16 = 80</math><br/><math>\therefore S_n</math> is true for <math>n=1</math><br/>Assume <math>S_n</math> is true for <math>n=k</math><br/><math>\therefore S_k: 13 \times 6^k + 2 = 5K</math><br/><math>\therefore S_{k+1} = 13 \times 6^{k+1} + 2</math><br/><math>= 13 \times 6^k \times 6 + 2</math><br/><math>= (13 \times 6^k)6 + 2</math><br/><math>= (5K - 2)6 + 2</math> (from <math>13 \times 6^k + 2 = 5K</math>)<br/><math>= 30K - 12 + 2</math><br/><math>= 30K - 10</math><br/><math>= 5(6K - 2)</math><br/><math>= 5M</math> where <math>M</math> is an integer,<br/>since <math>K</math> is a positive integer<br/><math>\therefore S_n</math> is true for <math>n=k+1</math><br/>Since <math>S_n</math> is true for <math>n=1</math>, it is true for <math>n=2</math>,<br/><math>\therefore</math> it is true for <math>n=3</math>, and so on<br/><math>\therefore S_n</math> is true for all positive integers.</p> | <p>1 mark awarded for partial correct solution</p> <p>2 marks awarded for further partial correct solution</p> <p>3 marks for complete correct solution</p> |
| P7                                                    | <p>(b) <math>\frac{d(xe^{\sin x})}{dx} = x \cdot e^{\sin x} \cdot \cos x + e^{\sin x} \cdot 1</math><br/><math>= e^{\sin x} (x \cos x + 1)</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>1 mark awarded for partial correct solution</p> <p>2 marks for complete correct solution</p>                                                             |
| H8                                                    | <p>(c) <math>A = \int_0^{\frac{\pi}{4}} \left( \sin 2x - \frac{4x}{\pi} \right) dx</math><br/><math>= \left[ \frac{-\cos 2x}{2} - \frac{4x^2}{2\pi} \right]_0^{\frac{\pi}{4}}</math><br/><math>= \left[ \frac{-\cos 2x}{2} - \frac{2x^2}{\pi} \right]_0^{\frac{\pi}{4}}</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>1 mark awarded for partial correct solution</p> <p>2 marks for complete correct solution</p>                                                             |

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|     | $= \left( \frac{-\cos 2\left(\frac{\pi}{4}\right)}{2} - \frac{2\left(\frac{\pi}{4}\right)^2}{\pi} \right) - \left( \frac{-1}{2} - 0 \right)$ $= 0 - \frac{\pi}{8} + \frac{1}{2}$ $= \frac{1}{2} - \frac{\pi}{8}$                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                             |
| H8  | <p>(d) <math>y = \log_e x, e^y = x, x^2 = e^{2y}</math><br/><math>V = \pi \int_0^6 x^2 dy</math><br/><math>= \pi \int_0^6 e^{2y} dy</math><br/><math>= \pi \left[ \frac{1}{2} e^{2y} \right]_0^6</math><br/><math>= \pi \left\{ \left( \frac{1}{2} e^{2(6)} \right) - \frac{1}{2} e^{2(0)} \right\}</math><br/><math>= \frac{1}{2} \pi (e^6 - 1)</math></p>                                                                                                                                                                                             | <p>1 mark awarded for partial correct solution</p> <p>2 marks for complete correct solution</p>                                                             |
| HE6 | <p>(e) <math>u = \sqrt{x} = x^{\frac{1}{2}}</math><br/><math>\frac{du}{dx} = \frac{1}{2} x^{-\frac{1}{2}}</math><br/><math>\frac{du}{dx} = \frac{1}{2\sqrt{x}}</math><br/><math>du = \frac{1}{2\sqrt{x}} dx</math><br/><math>2du = \frac{1}{\sqrt{x}} dx</math></p> <p>When<br/><math>x=0, u=0</math><br/><math>x=3, u=\sqrt{3}</math></p> $\therefore \int_0^3 \frac{dx}{\sqrt{x}(1+x)} = \int_0^{\sqrt{3}} \frac{2du}{1+u^2}$ $= 2 [\tan^{-1} u]_0^{\sqrt{3}}$ $= 2 (\tan^{-1} \sqrt{3} - \tan^{-1} 0)$ $= 2 \times \frac{\pi}{3}$ $= \frac{2\pi}{3}$ | <p>1 mark awarded for partial correct solution</p> <p>2 marks awarded for further partial correct solution</p> <p>3 marks for complete correct solution</p> |

| Year 12 Mathematics Extension 1<br>Question No. 6 Solutions and Marking Guidelines<br>Outcomes Addressed in this Question:<br>HE4 uses the relationship between functions, inverse functions and their derivatives |                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                       |
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| Outcome                                                                                                                                                                                                            | Solutions                                                                                                                                                                                                                                                                                                    | Marking Guidelines                                                                                                                                                                                                                                                                                                    |
| HE4 (a)                                                                                                                                                                                                            | $\int_0^1 \frac{dx}{\sqrt{2-x^2}} = \sin^{-1} \frac{x}{\sqrt{2}} \Big _0^1 = \frac{\pi}{4}$                                                                                                                                                                                                                  | <u>2 marks</u> <ul style="list-style-type: none"> <li>Correct solution</li> </ul> <u>1 mark</u> <ul style="list-style-type: none"> <li>Substantial progress.</li> </ul>                                                                                                                                               |
| (b)                                                                                                                                                                                                                | Domain: $-\frac{1}{3} \leq x \leq \frac{1}{3}$<br>Range: $-\pi \leq y \leq \pi$<br>                                                                                                                                         | <u>3 marks</u> <ul style="list-style-type: none"> <li>Correct solution (All 3 components correct)</li> </ul> <u>2 marks</u> <ul style="list-style-type: none"> <li>Substantial progress towards solution</li> </ul> <u>1 mark</u> <ul style="list-style-type: none"> <li>Partial progress towards solution</li> </ul> |
| (c)                                                                                                                                                                                                                | $\frac{dy}{dx} = \frac{-1}{\sqrt{1 - \left(\frac{1}{x}\right)^2}} \times \frac{-1}{x^2}$ $= \frac{1}{x^2 \sqrt{\frac{x^2-1}{x^2}}}$ $= \frac{1}{x\sqrt{x^2-1}}$                                                                                                                                              | <u>2 marks</u> <ul style="list-style-type: none"> <li>Correct solution</li> </ul> <u>1 mark</u> <ul style="list-style-type: none"> <li>Substantial progress.</li> </ul>                                                                                                                                               |
| (d)                                                                                                                                                                                                                | (i) $f'(x) = -1(e^{-x} + e^x) = -\left(\frac{1}{e^x} + e^x\right)$<br>As $e^x > 0$ for all values of $x$ , then the expression inside the brackets is always positive. Multiplication by -1 gives a negative derivative for all values of $x$ . This means that $f(x)$ is decreasing for all values of $x$ . | <u>1 mark</u> <ul style="list-style-type: none"> <li>Correct derivative as well as indicating the negative derivative for all values of <math>x</math>.</li> </ul>                                                                                                                                                    |

|                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| (ii) $f^{-1}(x): x = e^{-y} - e^y = \frac{1}{e^y} - e^y$<br>$\rightarrow e^{2y} + xe^y - 1 = 0$<br>Using quadratic formula: $e^y = \frac{-x \pm \sqrt{x^2 + 4}}{2}$<br>For all values of $x$ , $x < \sqrt{x^2 + 4}$ , $\therefore -x + \sqrt{x^2 + 4} > 0$<br>But $-x - \sqrt{x^2 + 4} < 0$ for all values of $x$<br>Therefore only solution is $e^y = \frac{-x + \sqrt{x^2 + 4}}{2}$<br>And given solution follows from this assertion. | <u>3 marks</u> <ul style="list-style-type: none"> <li>Correct solution, including justification of positive and negative values of <math>e^y</math>. (You can't just assume that 1 solution will be negative. It's not always true for a quadratic.)</li> </ul> <u>2 marks</u> <ul style="list-style-type: none"> <li>Substantial progress towards solution</li> </ul> <u>1 mark</u> <ul style="list-style-type: none"> <li>Partial progress towards solution</li> </ul> |
| (iii) Solution is found by substituting $x = 6$ into the solution given for $f^{-1}(x)$ in part (ii).<br>Calculator answer = $-1.82$                                                                                                                                                                                                                                                                                                     | <u>1 mark</u> <ul style="list-style-type: none"> <li>Correct solution.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                        |

| Year 12 Mathematics Extension 1 Task 4 2009 Trial HSC                                            |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                   |
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| Question No. 7 Solutions and Marking Guidelines                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                   |
| Outcomes Addressed in this Question                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                   |
| H5: applies appropriate techniques from the study of geometry and trigonometry to solve problems |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                   |
| H8: uses techniques of integration to calculate areas and volumes                                |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                   |
| Outcome                                                                                          | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                | Marking Guidelines                                                                                                                                                                                                |
| Q7(a)(i)<br><br>HS                                                                               | $\cos 2x = 2 \cos^2 x - 1$ $\therefore \cos^2 x = \frac{1}{2}(\cos 2x + 1)$ $\int_0^{\frac{\pi}{4}} \cos^2 x \, dx = \frac{1}{2} \int_0^{\frac{\pi}{4}} (\cos 2x + 1) \, dx$ $= \frac{1}{2} \left[ \frac{1}{2} \sin 2x + x \right]_0^{\frac{\pi}{4}}$ $= \frac{1}{2} \left\{ \frac{1}{2} \sin \frac{\pi}{2} + \frac{\pi}{4} \right\}$ $= \frac{1}{2} \left\{ \frac{1}{2} + \frac{\pi}{4} \right\}$ $= \frac{\pi + 2}{8}$ | 2 marks: Fully correct solution<br>1 mark: Uses double angle result for $\cos 2x$ and integrates.                                                                                                                 |
| Q7(a)(ii)<br><br>H8                                                                              | $y = \cos x + \sec x$ $y^2 = \cos^2 x + \sec^2 x + 2$ $V = \pi \int_0^{\frac{\pi}{4}} y^2 \, dx$ $= \pi \int_0^{\frac{\pi}{4}} (\cos^2 x + \sec^2 x + 2) \, dx$ $= \pi \left[ \frac{\pi + 2}{8} + \pi \left[ \tan x + 2x \right]_0^{\frac{\pi}{4}} \right]$ $= \pi \left\{ \frac{\pi + 2}{8} + 1 + \frac{\pi}{2} \right\}$ $= \pi \left\{ \frac{\pi + 2 + 8 + 4\pi}{8} \right\}$ $= \frac{5\pi(\pi + 2)}{8}$             | 3 marks<br>Fully correct solution.<br>2 marks<br>Obtains correct simplified expression for volume before substitution of limits<br>1 mark<br>Obtains correct simplified expression for volume before integration. |
| Q7(b)(i)<br>HS                                                                                   | $\text{In } \triangle TOM, \sin \theta = \frac{OM}{l}$ $OM = l \sin \theta$ $\text{Similarly, } MT = l \cos \theta$                                                                                                                                                                                                                                                                                                      | 2 marks: Correct results for both OM and MT.<br>1 mark: Correct result for one of OM or MT                                                                                                                        |
| (ii)                                                                                             | $\text{In } \triangle PTM, \tan \alpha = \frac{TM}{PM}$ $PM = \frac{l \cos \theta}{\tan \alpha}$ $= l \cos \theta \cot \alpha$                                                                                                                                                                                                                                                                                           | 2 marks: Justification of result.<br>1 mark: Sufficient progress towards result                                                                                                                                   |

|       |                                                                                                                                                                                                                                                                                                              |                                                                                 |
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| (iii) | Now in $\triangle OPM$ , $PM^2 = OP^2 + OM^2$<br>$l^2 \cos^2 \theta \cot^2 \alpha = d^2 + l^2 \sin^2 \theta$ $l^2 \cos^2 \theta \cot^2 \alpha - l^2 \sin^2 \theta = d^2$ $l^2 \{ \cos^2 \theta \cot^2 \alpha - \sin^2 \theta \} = d^2$ $l^2 = \frac{d^2}{\{ \cos^2 \theta \cot^2 \alpha - \sin^2 \theta \}}$ | 2 marks: Justification of result.<br>1 mark: Sufficient progress towards result |
| (iv)  | $l^2 = \frac{25^2}{\{ \cos^2 20^\circ \cot^2 24^\circ - \sin^2 20^\circ \}}$ $l = 12m$                                                                                                                                                                                                                       | 1 mark: Correct solution                                                        |